

Enhancing Navigation for Railway Station Facilities and Locations

Project Synopsis

By

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Introduction:

Railway stations are bustling and often confusing spaces with numerous facilities such as ticket counters, platforms, restrooms, food courts, waiting rooms, and exits. For first-time passengers, tourists, elderly travelers, and people with disabilities, navigating through these complex environments can be stressful and time-consuming. A smart, user-friendly navigation system is essential to improve the efficiency and quality of travel experiences in railway stations.

Project Category:

Category: Application

Hardware components: Laptop, Mobile

Software components: Visual Studio Code

Programming Languages : Java Scripts, HTML/CSS, APIs, Mapping Libraries

Objective of the project:

To develop a multi-platform navigation system that provides passengers with real-time, interactive, and accessible directions to all facilities and services within a railway station, ensuring a smoother and more informed travel experience.

System Requirements:

Hardware:

Operating System: Windows 7/8/8.1/10

RAM: 4GB

Hard Disk: 500MB

Processor: Intel Multi Core or High Processor

Software:

Visual Studio Code

APIs

Mapping Libraries

Scope:

Maps showing platforms, restrooms, counters, exits, shops, etc.

Turn-by-turn walking direction

Daily commuters for quick and easy movement

First-time travelers/tourists who need help navigating unfamiliar stations

Senior citizens and people with disabilities needing accessible paths

Station staff for managing updates and assisting passengers

Railway authorities for better crowd flow and service management

Result:

The proposed system was successfully designed and prototyped to enhance the navigation experience within railway stations. The final result is a multi-platform smart navigation system that allows passengers to easily locate and access station facilities such as platforms, ticket counters, restrooms, food courts, and waiting areas.